Summary

Port Alder is a coastal city of 410,000 people that recorded 31 days above 35 degrees Celsius in the summer of 2029, more than double the average of the previous decade. This study reports on the first five years of the city's heat adaptation programme, which ran from 2029 to 2033 and focused on cool roofs, tree planting and neighbourhood cooling centres.

The headline result is that heat-related emergency room visits fell by 22 percent over the study period even as average summer temperatures kept rising. The authors attribute most of that fall to the cooling centres and the outreach around them, not to the physical works alone.

The study compares twelve treated neighbourhoods with twelve matched neighbourhoods that received the programme two years later. Data came from hospital records, indoor temperature loggers in 1,800 homes and yearly household interviews.

The programme cost 58 million dollars in total. Two thirds came from a city climate fund financed by a hotel tax, and one third came from a national health grant. No private capital was involved, which the authors say kept the programme focused on the poorest districts.

Cool roofs

Reflective coatings were applied to 9,400 homes, prioritising single-storey housing with metal roofs. Across the logged homes, cool roofs lowered average indoor temperature during heatwaves by 3.1 degrees Celsius. The benefit was largest on top floors and small on lower floors of taller buildings.

Coatings lose reflectivity as they weather. The study measured about a third of the benefit lost after four years without cleaning, so it recommends budgeting for recoating every five years.

Households rarely complained about appearance but often asked whether coatings would reduce heating comfort in winter. The loggers showed a winter penalty of under half a degree, which the authors consider small in Port Alder's mild climate and possibly significant in colder places.

Tree canopy

The city planted 61,000 street trees with the goal of raising canopy cover from 18 percent to 27 percent by 2045. Canopy reached 20 percent by 2033. Trees grow slowly, so this measure had almost no effect on health results during the study, and the authors warn against judging it on a five-year window.

About 14 percent of new trees died in their first two summers. Survival rose sharply where residents were paid small stipends to water them, which became a standard part of the programme in 2031.

Cooling centres and outreach

Fourteen neighbourhood cooling centres opened in libraries, schools and community halls, staying open until 10 p.m. during heat alerts. On their own they drew few visitors in the first summer. Attendance tripled after the city hired 120 community ambassadors, mostly older residents, to phone and visit neighbours at risk.

The ambassadors programme cost about 1.4 million dollars a year. The authors rank it as the most cost-effective part of the whole programme and estimate that it accounts for more than half of the fall in emergency visits.

Who benefited

Emergency visits fell fastest among people over 65 living alone, by 34 percent, and slowest among outdoor workers, by only 6 percent. Outdoor workers were not reached by centres or roofs, and the authors recommend a separate workplace rule on rest breaks during heat alerts.

Renters benefited less than owners from roof coatings because landlords had to agree to the works, and about one in five refused.

Limits of the study

The comparison neighbourhoods were not randomly chosen, and the COVID-era decline in hospital use may affect the early baseline. The study does not measure deaths, because numbers were too small to separate from chance. It also does not cover flooding or other coastal hazards, which the city handles under a different department.

Future work should follow the delayed neighbourhoods and check whether benefits last once programme funding ends in 2036.